



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

PAPER 1 Multiple Choice

4025/1

1 hour

JUNE 2026 SESSION

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

Calculator (Optional)

BYWb

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

Answer **all** questions. For each question, there are **four** possible answers, **A, B, C** and **D**. Choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet provided.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

This question paper consists of 13 printed pages.

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- 1 Which branch of Biology is concerned with the study of function?
- A anatomy
 - B botany
 - C physiology
 - D genetics

- 2 Which food nutrient has the highest energy content?
- A carbohydrates
 - B lipids
 - C proteins
 - D nucleic acids

- 3 Glycerol can be obtained from the breakdown of
- A starch.
 - B proteins.
 - C nucleic acids.
 - D lipids.

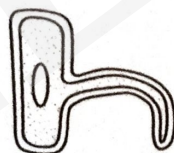
- 4 Which reagents are used to test for protein?
- A absolute ethanol and a hydroxide
 - B iodine solution and Benedict's solution
 - C Benedict's solution and hydrochloric acid
 - D copper sulphate solution and a hydroxide

- 5 The diagram shows four specialised cells.

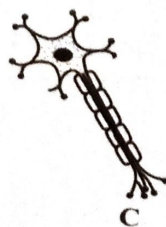
Which cell, A, B, C or D, conducts electrical impulses?



A



B



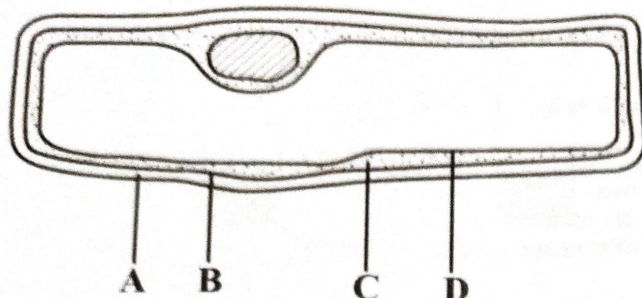
C



D



- 6 The diagram shows the structure of a palisade cell.
Which part, **A**, **B**, **C** or **D**, provides mechanical support?



- 7 Which one best describes an enzyme?
- A** A biological catalyst that acts a chemical reactant.
 - B** A carbohydrate that acts as a biological catalyst.
 - C** A protein that acts as a biological catalyst.
 - D** A biological molecule that acts as a substrate.
- 8 A student added iodine to a mixture of starch and amylase. The blue black colour disappeared after some time.
The change in colour is due to the formation of
- A** sucrose.
 - B** glucose.
 - C** fructose.
 - D** maltose.
- 9 The mode of action of enzymes can be explained using the lock and key hypothesis.
Which row correctly identifies the enzyme and substrate?

	active site is on the enzyme	active site on the substrate	enzyme is the lock	substrate is the key
A	×	✓	✓	×
B	×	✓	×	✓
C	✓	×	✓	✓
B	✓	×	×	✓

Key:

present ✓

absent ×

- 10 Crop plants with yellow and brown leaf margins need a fertilizer rich in
- A calcium.
 - B nitrogen.
 - C phosphorus.
 - D potassium.

- 11 Plant leaves minimize water loss by
- A developing hairs.
 - B increasing surface area.
 - C reducing thickness of cuticle.
 - D increasing number of stomata.

- 12 The diagram shows a plant leaf damaged by a pest.

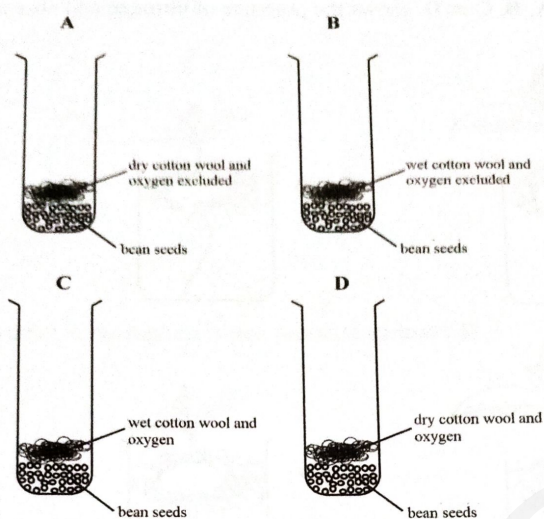


Which pest might have caused the damage?

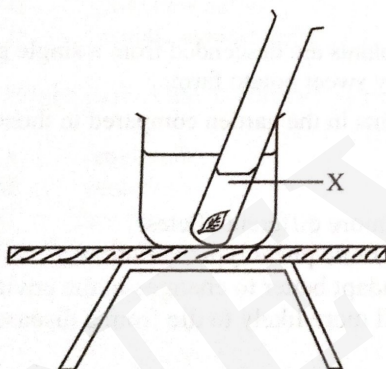
- A aphid
 - B locust
 - C earth worm
 - D red spider mite
- 13 Which set of plant hormones are added to cuttings to promote root and shoot development?
- A auxins and gibberellins
 - B cytokinin and ethylene
 - C ethylene and gibberellins
 - D auxins and cytokinin
- 14 Which statement correctly describes the changes that occur in a flower after fertilisation?
- A Ovule develops into a seed and ovary into a fruit.
 - B Ovule develops into a fruit and ovary into a seed.
 - C Petals remain the same and ovary develops into a fruit.
 - D Ovule develops into a seed and stigma into sepals.



- 15 An experiment was carried out to investigate germination.
In which test tube, A, B, C or D, is germination likely to occur?



- 16 The diagram shows a stage in testing a leaf for starch. The test tube with the leaf was placed in a hot water bath.

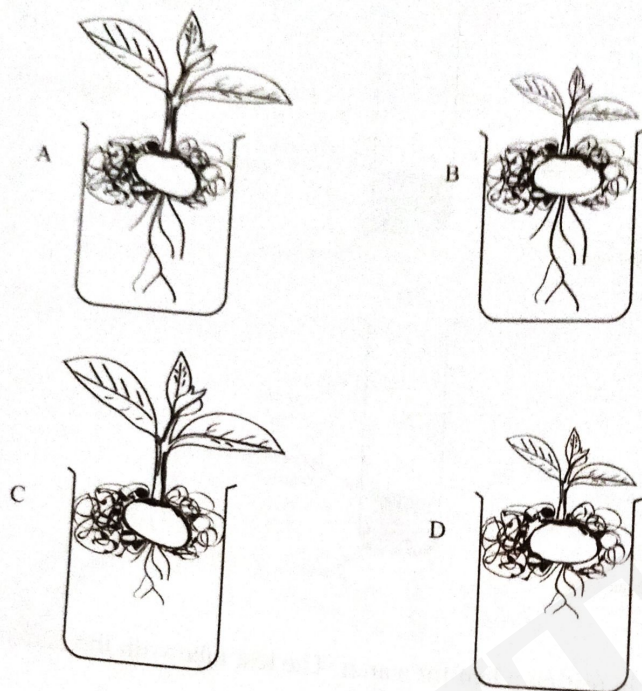


What is the purpose of liquid X?

- A to remove chlorophyll
- B to soften the leaf
- C to remove sugars
- D to stop chemical reactions



- 17 The diagrams show four different plants of the same type.
Which plant, A, B, C or D, shows the presence of nitrogen and absence of phosphorus?



- 18 In a garden, the entire population of sweet potato plants are descended from a single parent. The parent potato plant was obtained from a nearby sweet potato farm.
Which statement is true about the sweet potato plants in the garden compared to those at the farm?

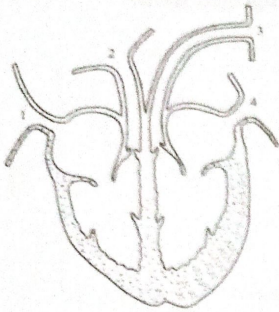
- A The sweet potato plants in the garden have more different alleles.
- B The sweet potato plants in the garden show more phenotypic variation.
- C The sweet potato plants in the garden will adapt better to changes in the environment.
- D The sweet potato plants in the garden are all more likely to die from a disease.

- 19 Which response will generate body heat?

- A shivering
- B sweating
- C vasodilation
- D vasoconstriction



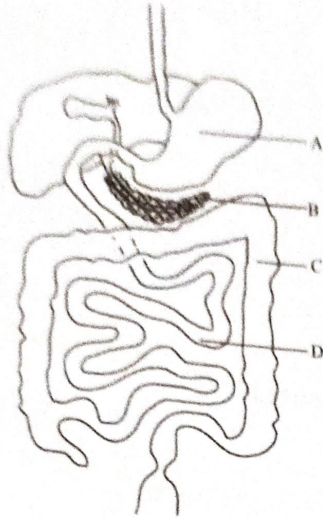
- 20 The diagram shows a mammalian heart.



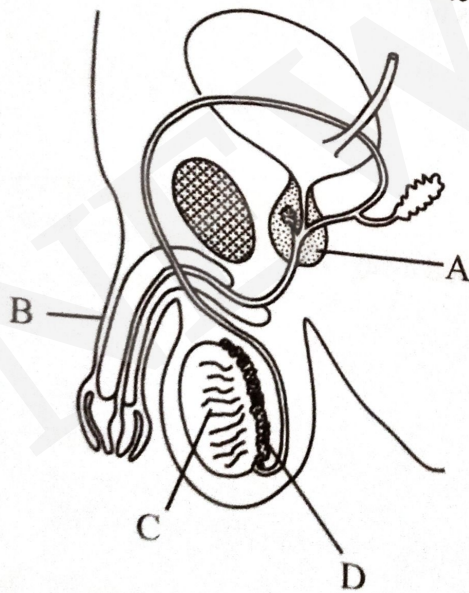
In which blood vessel is the highest blood pressure measured?

- A 1
B 2
C 3
D 4
- 21 The major function of carbohydrates in a diet is to
- A fight infection.
B supply energy.
C grow new cells.
D insulate against cold.
- 22 Which part of the circulatory system carries blood away from the ileum?
- A renal vein
B vena cava
C hepatic vein
D portal vein

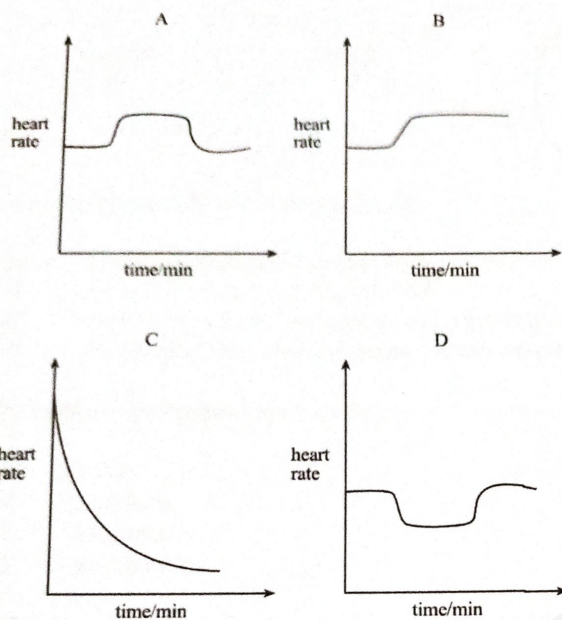
- 23 The diagram shows the human alimentary canal.
Which part, **A**, **B**, **C** or **D**, is adapted for the absorption of digested food?



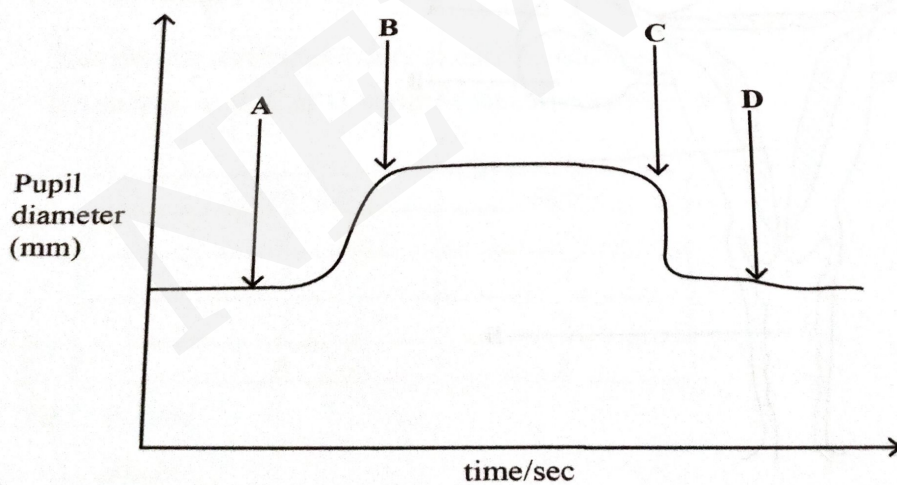
- 24 The diagram shows the male reproductive system.
Which part, **A**, **B**, **C** or **D**, produces testosterone?



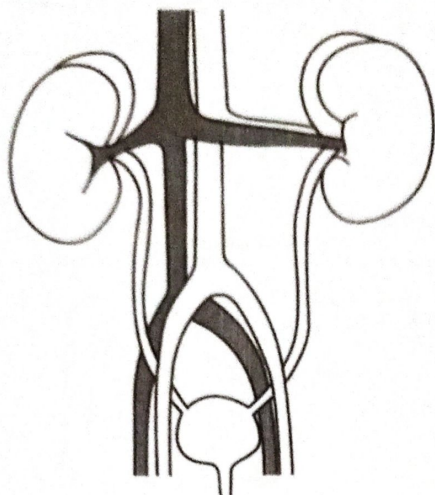
- 25 Which graph, **A**, **B**, **C** or **D**, correctly shows a person's heart rate before, during, and after a period of exercise?



- 26 The graph shows changes in pupil size at different light intensities. At which point, **A**, **B**, **C** or **D**, does light intensity increase?



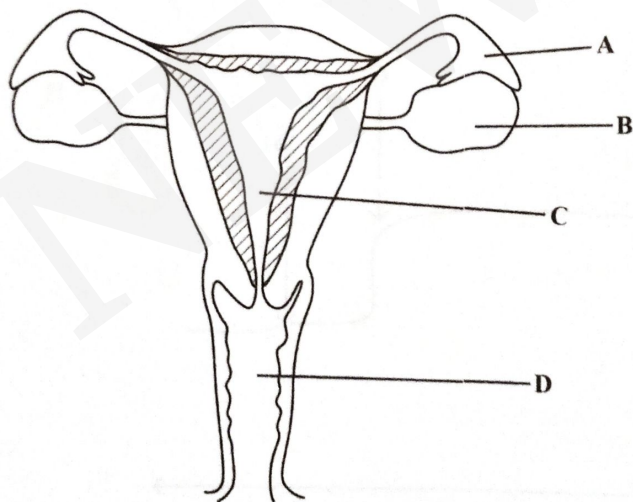
- 27 The diagram shows some internal organs of the human body.



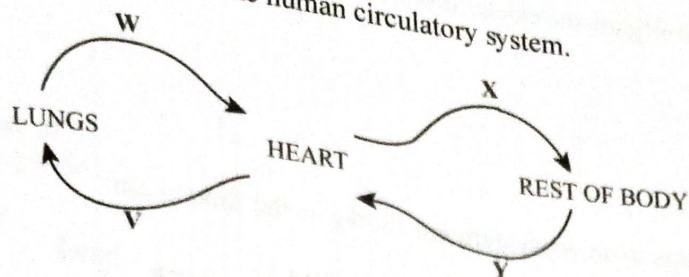
The organs make up the

- A reproductive system.
- B digestive system.
- C respiratory system.
- D urinary system.

- 28 The diagram shows the female reproductive system.
In which part, A, B, C or D, does implantation occur?



- 29 The diagram illustrates the human circulatory system.



Arrows labelled X and Y represent the

- A vena cava and aorta respectively.
- B aorta and vena cava respectively.
- C pulmonary artery and pulmonary vein respectively.
- D pulmonary vein and pulmonary artery respectively.

- 30 Hormones are broken down in the

- A liver.
- B stomach.
- C kidneys.
- D duodenum.

- 31 Which one is a benefit of recombinant DNA technology?

- A changing the environment
- B changing future generations
- C increased genetic diversity
- D increased yield in Agriculture

- 32 The table shows characteristics of micro-organisms.
Which row, A, B, C or D, correctly identifies a bacterium?

	unicellular	capsid
A	✓	✓
B	x	✓
C	✓	x
D	x	x

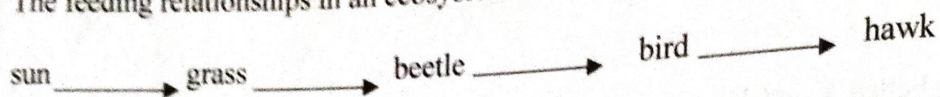
key ✓ present

x absent

33 Which group of micro-organisms causes diseases which **cannot** be cured using drugs?

- A bacteria
- B protozoa
- C fungi
- D virus

34 The feeding relationships in an ecosystem are shown as the food chain:



Which link has the least amount of energy transferred?

- A sun → grass
- B grass → beetle
- C beetle → bird
- D bird → hawk

35 Drugs taken to prevent malaria are

- A anaesthetics.
- B prophylactics.
- C analgesics.
- D sedatives.

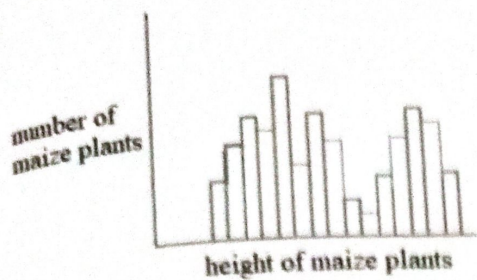
36 Malaria is caused by

- A mosquitoes.
- B plasmodia.
- C bacteria.
- D viruses.

37 What is a mutation?

- A Appearance due to recessive alleles.
- B A change in a gene or a chromosome.
- C Changing genes by recombinant DNA technology.
- D Variation shown by individuals of the same species.

The graph shows the height of maize plants growing in a field.



Which type of variation is shown by the maize plants?

- A continuous variation
- B discrete variation
- C continuous and discontinuous variation
- D neither continuous nor discontinuous variation

39 Which term describes the variety of life in particular area?

- A biodiversity
- B ecosystem
- C fauna
- D flora

40 Which one is a unique characteristic of the kingdom *Plantae*?

- A respire
- B photosynthesize
- C saprophytic
- D disperse

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NEWTON

